

Verification of Effectiveness of a Threshold-Based Jerk Penalty Term for Reducing Occupant Sway During Stair Climbing in Physical Quadruped Wheelchairs

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Abstract—We propose a reinforcement learning method using a threshold-based jerk penalty term with the aim of reducing occupant vibration during stair climbing and descending on a physical quadruped wheelchair. A mannequin simulating a chair and an occupant was mounted on a Unitree Go2, and vibration was measured using three inertial measurement units (IMUs) installed on the head and both hands. We compared four conditions—Baseline (standard reward term for Unitree Go2), Acceleration-Tilt Comfort, Head Jerk Comfort, and Multi-point Jerk Comfort—and evaluated power spectral density (PSD) integral values in the 0–10 Hz range for ten successful trials each of ascending and descending stairs. The results showed that the Head Jerk Comfort condition, which targeted only the head, exhibited the lowest PSD integral values at all measurement points and demonstrated the most consistent vibration reduction effect for both ascending and descending. Accordingly, we confirmed that the threshold-based jerk penalty term is an effective reward design for reducing occupant vibration while maintaining stair-climbing performance.

I. INTRODUCTION

Legged robots possess excellent off-road mobility, and their application to wheelchairs is expected to provide mobility assistance in environments that include steps and stairs [1–4]. However, since vibrations are transmitted to the occupant due to ground impact and changes in posture during legged walking, control strategies must account for not only mobility performance but also occupant comfort. In recent years, deep reinforcement learning has been widely used for the gait control of legged robots [5–8]; however, much of the existing research focuses primarily on speed tracking and posture stability, whereas reward designs that directly reduce occupant vibration have not been sufficiently explored. In particular, if jerk—which is the time derivative of acceleration—is used directly as a penalty, it may suppress even the large movements necessary for ascending and descending stairs, potentially hindering gait learning. To deal with this issue, in this study, we propose a threshold-based jerk penalty term that uses threshold processing to separate large body movements from micro-vibrations. In experiments using a real-world four-legged wheelchair based on the Unitree Go2, we compared four conditions—Baseline (standard reward term for Unitree Go2), Acceleration-Tilt Comfort, Head Jerk Comfort, and Multi-point Jerk Comfort—and evaluated them by performing a power spectral density (PSD) analysis of inertial measurement units (IMU) data from the head and both hands. The main contributions of this study are: (1) proposal of a threshold-based jerk penalty term aimed at reducing occupant

vibration in a real-world four-legged wheelchair; (2) quantitative evaluation using three-point IMU data and PSD integral values during real-world stair ascents and descents; and (3) demonstration that a jerk penalty targeting only the head exhibits the most consistent vibration reduction effect among the four conditions.

II. RELATED WORK

To date, the authors have investigated stair climbing using a quadruped wheelchair [1,2], the acquisition and evaluation of walking and wheeled mobility policies using reinforcement learning [3], and the integration of walking and wheeled mobility via mode-conditional policy distillation [4]. These studies have primarily focused on traversal performance and the acquisition and integration of locomotion modes; however, no research has been conducted to directly reduce occupant vibration during stair climbing on an actual vehicle through reward design. Regarding vibration reduction in legged robots, studies have reported vibration suppression in the head of a quadruped robot [9] and tilt suppression of carried objects [10]. Furthermore, for wheeled-legged robots, motion control combining walking and wheeled driving has also been investigated [11]. On the other hand, while evaluation methods and comfort metrics for whole-body human vibration have been systematized in ISO 2631-1 [12] and by Griffin [16], there are few studies that have directly incorporated these into the reinforcement learning rewards for legged wheelchairs. Furthermore, deep reinforcement learning is widely used for acquiring walking policies in legged robots [5–8]. On the basis of these findings, in this study, we introduce a jerk reward term that distinguishes between large-scale body movements and micro-vibrations using threshold processing and verify its effectiveness in reducing occupant sway in a physical quadruped wheelchair.

III. EXPERIMENTAL SETTINGS

A. Physical Quadruped Wheelchair

Figure 1 shows the physical quadruped wheelchair used in the experiment. A child’s seat was secured to the top of the Unitree Go2 using 3D-printed brackets, and a mannequin simulating an occupant was seated in it.

The mannequin was secured to the chair with a seatbelt. Meanwhile, to reproduce the passive swaying motion of an actual occupant’s head, the neck joint was loosely adjusted. The neck was lightly secured with tape to prevent the head

from falling off, but care was taken not to restrict the head's movement.

To measure the occupant's swaying motion, IMUs were attached to three locations—the head, the left wrist, and the right wrist—to acquire acceleration data.



Fig. 1. Physical quadruped wheelchair.

B. Staircase Test Environment

Figure 2 shows the experimental setup. The staircase was constructed using an aluminum frame and consisted of five steps, each with a riser height of 7 cm. Although this is smaller than the stairs typically found in buildings, it was designed to match the scale of the Unitree Go2, a small quadruped robot.

In the real-world experiments, to ensure safety, a rail was installed at the top of the stairs, and a passive fall prevention mechanism using a wire was implemented. This mechanism was adjusted so that no tension is applied during normal walking; thus, it did not affect the robot's movement and prevented damage to the robot only in the event of a fall.

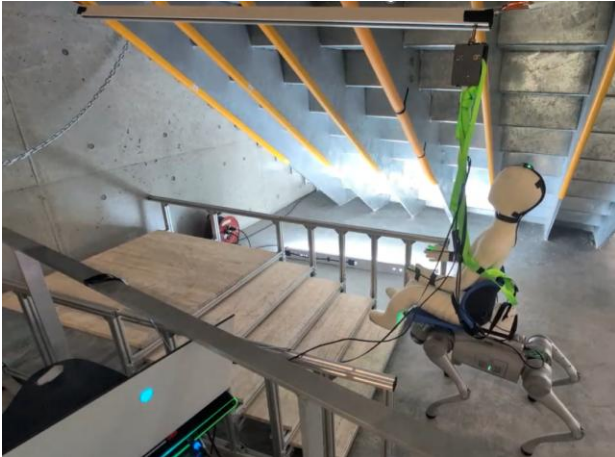


Fig. 2. Experimental environment.

C. Simulation Environment

Isaac Lab [8] was used for reinforcement learning. The simulation model consisted of Unitree Go2, the upper seat, and the occupant model, just as in the actual device. The total mass of the occupant model was set to 2 kg, the same as the mannequin used in the experiment, and the mass was distributed among the head, torso, upper limbs, and lower limbs based on previous studies [13,14] regarding the mass ratios of various body parts. For training, we used the terrain generation feature provided as standard by Isaac Lab and conducted reinforcement learning in multiple environments,

including stairs, ramps, and uneven terrain. Figure 3 shows a quadruped wheelchair equipped with the occupant model undergoing reinforcement learning within the simulation environment.

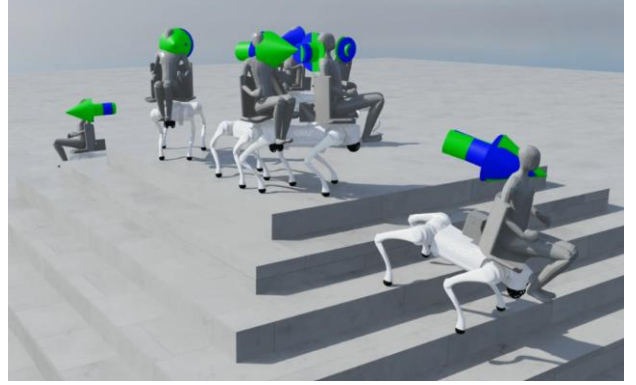


Fig. 3. Reinforcement learning in a simulation.

IV. REWARD DESIGN IN REINFORCEMENT LEARNING

We compared four types of reward design to verify the effectiveness of the proposed threshold-based jerk penalty term: Baseline, Acceleration-Tilt Comfort, Head Jerk Comfort, and Multi-point Jerk Comfort.

A. Baseline

As for the Baseline design, we used the standard reward function provided by Isaac Lab for Unitree Go2 as-is. We will refer to this as the baseline reward r_{base} .

The baseline reward includes reward terms commonly used in gait control, such as speed tracking, posture maintenance, torque control, and joint acceleration control. Due to space constraints, we will not describe the baseline in detail. Readers can consult [17] for details.

B. Acceleration-Tilt Comfort

Previous studies have proposed methods for suppressing changes in acceleration and the attitude of payloads [9,10]. Furthermore, in evaluations of whole-body vibration in humans, acceleration and frequency characteristics have been used to assess comfort [12,16]. We chose the acceleration-tilt comfort method as a comparative approach based on these studies. The reward function is as follows:

$$r_{acc_tilt} = r_{base} + r_{acc} + r_{tilt}, \quad (1)$$

where $r_{acc} = -0.005a^{(head)} - 0.0025(a^{(left)} + a^{(right)})$, and $r_{tilt} = -0.1t^{(chair)}$. Here, $a^{(head)}$, $a^{(left)}$, $a^{(right)}$ represent the accelerations measured by the IMUs on the head, left hand, and right hand, respectively. $t^{(chair)}$ represents the angle of inclination of the seat.

C. Head Jerk Comfort

We focused on the head—which is considered to have the greatest impact on occupant comfort—and introduced a new threshold-based jerk penalty term.

Generally, if jerk is incorporated directly into the reward, even large body movements that occur when starting to walk or ascending and descending stairs are suppressed, thereby hindering gait learning. Therefore, since this study targets only fine vibrations, we used the following equation:

$$r_{thr_jerk}^{(b)} = \left(\frac{\max(m_b - \theta, 0)}{s} \right)^2, \quad (2)$$

where m_b is the L2 norm of the jerk at location b . Furthermore, $\theta=1000$ and $s=1000$. The reward function of Head Jerk Comfort is as follows:

$$r_{head_jerk} = r_{acc_tilt} - 0.02r_{thr_jerk}^{(head)}. \quad (3)$$

In preliminary experiments, we confirmed that when jerk was used directly as a penalty without thresholding, the walking motion itself was suppressed. We then examined the distribution of jerk during walking and observed a large peak in the region of low jerk values. Although the individual values of these components were small, they occurred continuously during walking; consequently, the cumulative penalty became large, and we hypothesized that this was hindering walking learning. Therefore, in this study, we excluded low jerk values from the penalty calculation and evaluated only jerk values exceeding a threshold. The threshold $\theta=1000$ was set based on the jerk distribution and the results of the preliminary experiments as the boundary that excludes low-jerk components while maintaining the walking motion. Furthermore, $s=1000$ was set empirically to prevent the penalty for jerk exceeding the threshold from becoming excessively large and dominating the overall reward.

D. Multi-point Jerk Comfort

Finally, we introduced a threshold-based jerk penalty term not only for the head but also for both hands.

The reward function is as follows:

$$\begin{aligned} r_{multi_point_jerk} = & r_{acc_tilt} \\ & - 0.01r_{thr_jerk}^{(head)} \\ & - 0.005r_{thr_jerk}^{(left)} \\ & - 0.005r_{thr_jerk}^{(right)}. \end{aligned} \quad (4)$$

Since the head is located farther from the torso and is therefore more prone to significant sway compared to the hands, and since this sway is believed to have a greater impact on occupant discomfort, the weight of the jerk term for the head was set to 0.01, and that for each hand to 0.005. It should be noted that this study did not verify whether these coefficients are optimal; this remains a topic for future research.

V. EVALUATION METHOD

We conducted 50,000-step reinforcement learning on the Isaac Lab platform for each reward design. After training, we applied each model to a physical quadruped wheelchair and collected ten successful trials each for ascending and descending. Note that this study did not perform a quantitative evaluation of stair-climbing performance itself; rather, after confirming that each model was capable of climbing stairs, the vibrations transmitted to the occupant were compared using PSD integral values. Acceleration data was acquired at approximately 100 Hz from IMUs attached to the occupant's head, left hand, and right hand, and PSD analysis was performed on the three-axis acceleration norm using the Welch method [15] (Hann window, maximum segment length of 256 samples, 50% overlap). Furthermore, by integrating each PSD over the 0–10 Hz range, we quantified the vibration energy

transmitted to the occupant. We used these PSD integral values as a comparative metric for the various reward designs. The range of 0–10 Hz was set as the evaluation range to focus on low-frequency vibrations associated with occupant motion.

VI. EXPERIMENTAL RESULTS

A. Learning Results

Figure 4 shows the learning curves for each model. For all four models, learning converged stably, and the models were able to ascend and descend stairs. We confirmed that basic walking performance could be maintained when the proposed reward term was introduced.



Fig. 4. Trends in average rewards.

When the strategies developed during the learning phase were applied to a physical quadruped wheelchair, all models were able to navigate stairs. In the following subsection, we compare the vibrations transmitted to the occupant by using IMU data.

B. PSD Analysis

Figure 5 shows the PSD analysis results for the head, left hand, and right hand during the upward and downward movements. Each result was calculated from the acceleration data of ten successful trials.

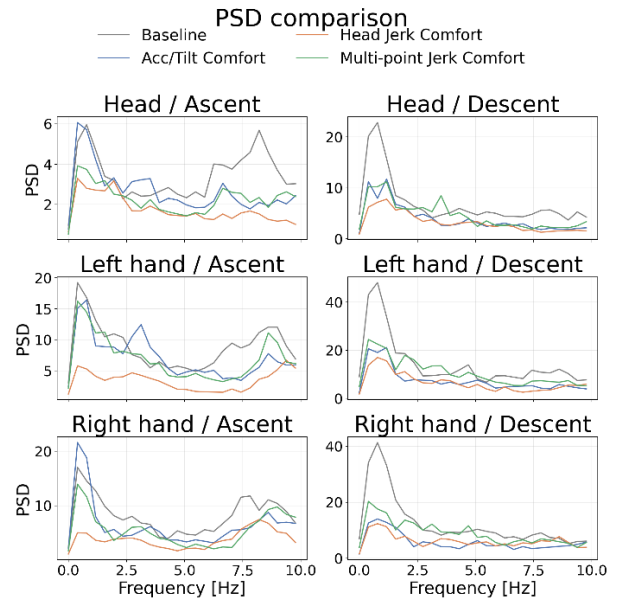


Fig. 5. PSD at each measurement point for ascent and descent motions.

During the ascent motion, “Head Jerk Comfort” exhibited the lowest PSD at all measurement points, confirming that it effectively suppressed vibrations. In particular, for the head, the PSD was reduced across a wider frequency band compared to the other three conditions. During the descent motion, although the differences between the models were not as significant as during the ascent motion, “Head Jerk Comfort” exhibited the lowest overall PSD.

C. Comparison Based on PSD Integral Values

Tables 1 and 2 show the results of integrals of the PSD over the range of 0 to 10 Hz for the ascent and descent motions.

TABLE I. INTEGRALS OF PSD VALUES DURING THE ASCENT MOTION

Model	Head	Left Hand	Right Hand
Baseline	34.93	88.08	83.41
Acc/Tilt Comfort	31.31	84.74	70.58
Head Jerk Comfort	19.58	33.82	40.96
Multi-point Jerk Comfort	24.23	73.36	57.36

TABLE II. INTEGRALS OF PSD VALUES DURING THE DESCENT MOTION

Model	Head	Left Hand	Right Hand
Baseline	74.47	158.66	135.39
Acc/Tilt Comfort	47.10	77.98	72.38
Head Jerk Comfort	37.37	71.25	71.04
Multi-point Jerk Comfort	50.81	118.03	101.83

During the ascent motion, Head Jerk Comfort exhibited the lowest values for the head, left hand, and right hand. Similarly, during the descent motion, Head Jerk Comfort also resulted in the lowest vibration energy.

On the other hand, for Multi-point Jerk Comfort, even though jerk penalty terms were introduced for the left and right hands in addition to the head, the results did not exceed those of Head Jerk Comfort.

One possible reason for this is that an optimization conflict may have arisen between the jerk penalty terms for the head and the left and right hands. In the Multi-point Jerk Comfort design, because the system attempted to reduce jerk at multiple body parts simultaneously, it learned a compromise strategy to balance the various objectives; as a result, it is believed that jerk reduction at any single body part did not exceed that achieved by the Head Jerk Comfort design. On the other hand, this study did not sufficiently verify the valid ranges and variability of jerk values for each body part or the appropriateness of the reward coefficients; these points remain topics for future investigation.

Overall, the above results indicate that the threshold-based jerk penalty function targeting only the head achieved the highest occupant comfort among the four reward designs compared in this study.

VII. CONCLUSION

We proposed a reinforcement learning method using a

threshold-based jerk penalty term with the aim of improving occupant comfort during stair climbing and descending in a real four-wheeled wheelchair. The proposed method seeks to balance walking performance and comfort by excluding jerks caused by large body movements through threshold processing and targeting only the minute vibrations that the occupant perceives as uncomfortable.

In the real-world evaluation, we compared four reward designs: Baseline, Acceleration-Tilt Comfort, Head Jerk Comfort, and Multi-point Jerk Comfort. The results confirmed that Head Jerk Comfort exhibited the lowest PSD integral values for both ascending and descending, demonstrating that it most effectively reduces the vibrations transmitted to the occupant. On the other hand, the Multi-point Jerk Comfort design—which introduced jerk penalty terms for the left and right fingertips in addition to the head—did not outperform Head Jerk Comfort, suggesting that simultaneously optimizing multiple body parts does not necessarily lead to improved comfort.

The results of the experiments confirmed that the proposed threshold-based jerk penalty term is effective for improving occupant comfort in quadruped wheelchairs. As for future work, we plan to analyze the factors contributing to the reduced performance of the Multi-point Jerk Comfort design and conduct further verification using diverse occupant conditions and more comprehensive comfort evaluation metrics.

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